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ex 5

Problem 1

Proof of Equivalence for tree Definitions

I will proof the equivalence by this cycle

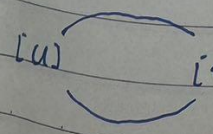
$1 \to 7 \to 6 \to 5 \to 4 \to 3 \to 2 \to 1$.

1. A tree is a connected acyclic graph.
2. A tree is one component of a forest.
3. A tree is a connected graph withte most $V-1$ edges
4. A tree is a minimally connected graph; removing any edge disconnects the graph
5. A tree is an acyclic graph with at lease $V-1$ edges
6. A tree is a maximally acyclic graph; adding an edge between any two vertices creates a cycle.
7. A tree is a graph that contains a unique path between each pair of vertices.

① $1 \to 7$

~~Stence~~ since the graph is connected, at least one path exists between any two vertices u and v.

If there were two path, their union would form a cycle.



cycle.

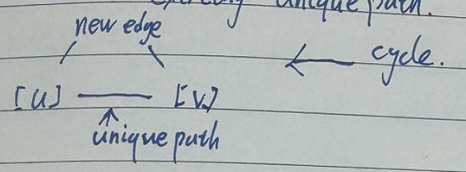
since the graph is acyclic, the path must be unique

WENZHANG

②. 7⇒6

Acyclic: If a cycle existed, there would be two paths between nodes on that cycle, contradicting Definition 7.

Maximally Acyclic: Adding an edge (u,v) creates a cycle with the existing unique path.



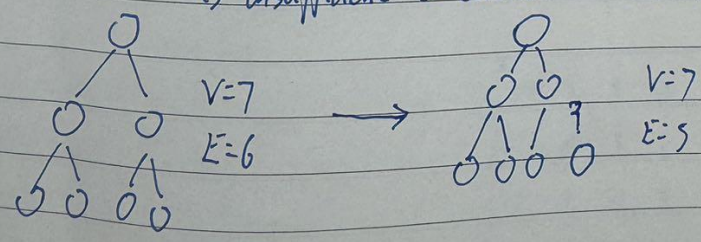
③ 6⇒5

For an acyclic graph, $E = V-k$. To be maximally acyclic, it must be connected ($k=1$), so $E = V-1$

④. 5⇒4

Connected: An acyclic graph with $E \geq V-1$ must be connected ($k=1$)

Minimally connected: Removing an edge ~~level~~ leaves $V-2$ edges, which is insufficient to connect V vertices.



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⑤ 4⇒3

If a connected graph had $E \geq V$, it would ~~be~~ contain a cycle. Removing a cycle edge would not disconnect it, contradicting D4.

⑥ 3⇒2

Acyclic: A connected graph with $V-1$ edges cannot be a cycle.
Forest Component: Being acyclic and connected makes it exactly one component of a forest.

⑦ 2⇒1

A forest is acyclic, and a component is connected. This yields D1.

Problem 2.

1. adjacence matrix reconstruction

vertex mapping: A=0, B=1, C=2, D=3, E=4

Col pointers: [0, 2, 5, 8, 11, 12]

V	Row	Neighbor
Col 0 (A)	1, 2	B, C
Col 1 (B)	0, 2, 3	A, C, D
Col 2 (C)	0, 1, 3	A, B, D
Col 3 (D)	1, 2, 4	B, C, E
Col 4 (E)	3	D.

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

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graph 2.

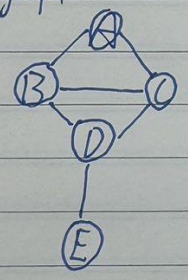
Col pointers: [0, 0, 2, 4, 5, 7]

V	Row	edges
Col 0 (A)	non	non
Col 1 (B)	0, 3	B→A, B→D
Col 2 (C)	0, 1	C→A, C→B
Col 3 (D)	2	D→C
Col 4 (E)	1, 3	E→B, E→D

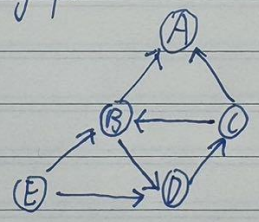
$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

2. Graph diagram

graph 1



graph 2.



3.

By tracing the edges in Graph 2, we can find a closed loop.

B → D → C → B